专利技术交底书

古地磁可靠性指标知识冲突检测方法

建议保护主线

✦ ✦ ✦ ✦ ✦

2026-07-17

核心结论： R3/R5/R7

一种基于原始文献、目标绑定证据关系和可撤销冲突推理的古地磁可靠性指标判定方法、系统、设备及存储介质

“ ”“ ” “R3 ”

1.
2.
3.
4.
5.
6.

“ ”

CN114881015B

CN118821777A《》

CN114266246B《》/

3.1

R3 R5 R7

- REFNO
- RESULTNO
- ROCKNAME

REFNO sibling target

3.2

REFNO

SHA-256

3.3

“ ” “ ”

3.4

SUPPORT CONFLICT UNKNOWN ACTIVE DEFEATED UNDECIDED
SUPPORTED_ONLY REFUTED_ONLY BOTH NEITHER 1 0 ABSTAIN

0 ABSTAIN

S1

R3 R5 R7

S2

S3

R3 R5 R7

S4

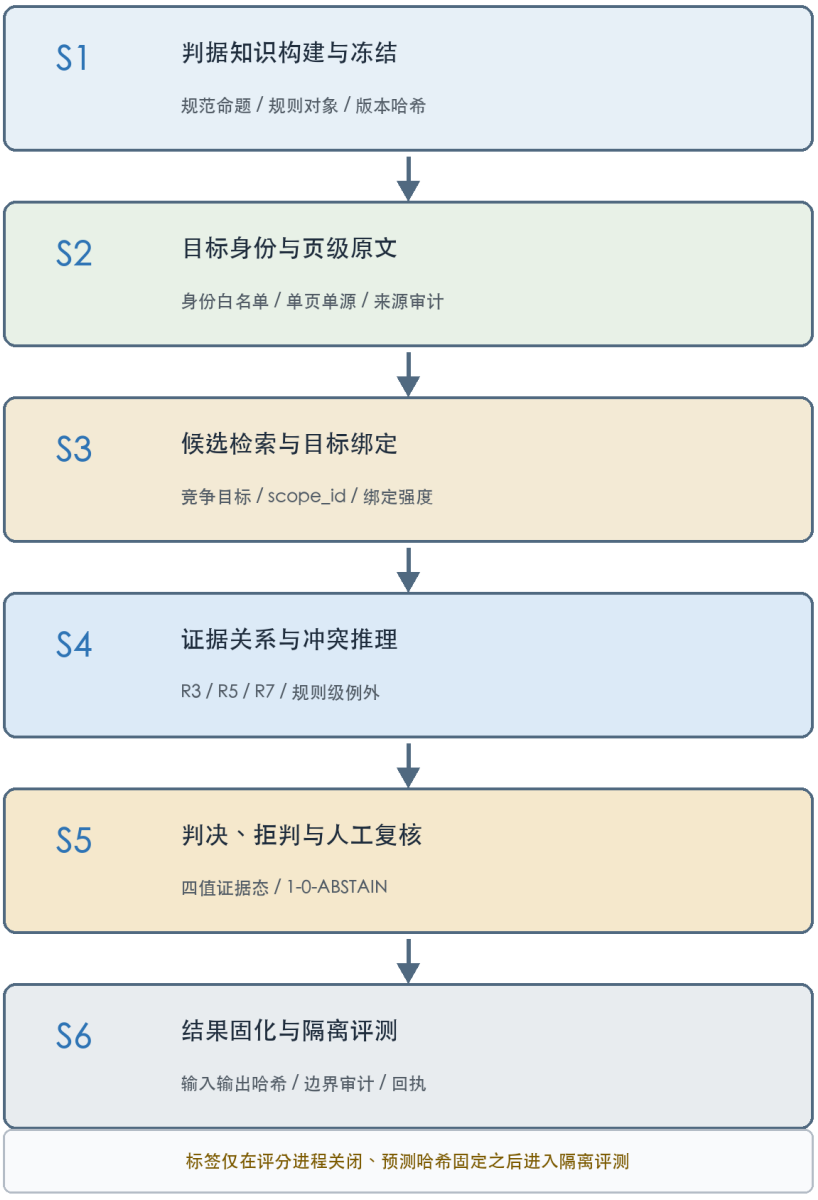
R3 R5 R7

S5

S6

SHA-256

古地磁可靠性指标知识冲突检测方法总体流程



S1

S11

R3 R5 R7 SHA-256

S12

`TRUE/FALSE/UNKNOWN` “ ”

S13

-
-
-
-
-

`all any not implies`

S14

`hematite haematite IRM acquisition SIRM Lowrie test Js-T FORC hysteresis`

S15

7.
8.
9.
10.
11.
12.
13.
14.

R3 0.48 R3 R5 0.78 R5 0.70 R7 0.82 R7 0.72

S16

R3 ROCKMAG `label_supervision=true` `REFNO` `RESULTNO` `label_supervision=false`

S17

`REFNO` `RESULTNO` gold label benchmark label ROCKMAG

S18

S1

S2

S21

`REFNO/RESULTNO/ROCKNAME` `RESULTNO` R1-R7 ROCKMAG

S22

`REFNO` PDF `SKIP_NO_SOURCE` 0 1

S23

PDF

S24

CID R3/R5/R7

S25

S26

-
-
- CID
-
-
- R3 -

-

S27

`selected_source` `rejected_source`

S28

`REFNO` `REFNO` `REFNO`

S29

PDF `UNKNOWN` `ABSTAIN`

S3

S31

R3 R5 R7
A95 R

S32

S33

`ROCKNAME` `REFNO` formation group basin granite

S34

scope_id scope_kind

S35

15.
16.
17.
18.
19.
20.

0 1 B_t

S36

scope_id

S37

REFNO + canonical ROCKNAME RESULTNO -
IDENTITY_AMBIGUOUS component manifest
RESULTNO

S38

```
C_e = clip(C_r × Q_s × (0.45 + 0.55 × B_t) × Q_m, 0, 1)
```

C_r Q_s B_t Q_m clip 0 1

S39

scope_id

S4

S41

`SUPPORT` `CONFLICT` `UNKNOWN`

S42

`ACTIVE` `UNDECIDED`
“ ”

S43 R3

R3

21.
22.
23. /

IRM FORC SEM R3 magnetite hematite pyrrhotite
`ABSTAIN`

S44 R3

“ ”“ ” “ ”

`criterion_practice_conflict=true`

S45 R5

R5

- “ ” “ ”
-
- R

tilt correction positive fold test undeformed small dip `post-folding secondary component` R5=0

S46 R5

“ ”
`DEFEATED` `ACTIVE`

S47 R7

A95 A95 R

```
d = arccos[sin(phi_t)sin(phi_c) + cos(phi_t)cos(phi_c)cos(lambda_t-lambda_c)]
```

`phi_t lambda_t` `phi_c lambda_c` `d <= A95_t + A95_c`

`R >= 3`

A95

S48 R7

“ ” R7 qualified-search manifest

S49

- `SUPPORTED_ONLY`
- `REFUTED_ONLY`
- `BOTH`
- `NEITHER`

S5

S51

`policy_decision=1` `policy_decision=0` `policy_decision=ABSTAIN`

S52

PDF `ABSTAIN` `compatibility_value`

S53

`RESULTNO` `IDENTITY_AMBIGUOUS` `ABSTAIN`

S54 BOTH

`BOTH` 0 `ABSTAIN` `REFUTED_ONLY`

S55

R3 `p <= 0.30` `p >= 0.70` `ABSTAIN`

S56

`ABSTAIN` `BOTH` -

S57

“ ”

JSON

S6

S61

R3 JSONL R5 R7 JSON

S62

HTML `ABSTAIN BOTH`

S63

SHA-256 R3/R5/R7 SHA-256

S64

S65

JSON JSONL JSON PMAGRESULT ROCKUNIT label evaluation lookup

S66

SHA-256

S67

`RESULTNO` `REFNO`

S68

macro-F1 F1(0) F1(1) grouped OOF

R3

24. `REFNO=P001 RESULTNO=R001 ROCKNAME=Target Formation Sibling Formation`

25. S2 PDF 12 13 12 13 13 `IRM` `magnetite` `remanence` 12 13
26. S3 13 IRM acquisition magnetite `Target Formation` `scope_id`
27. R3

28. S4 `scope_id` `SUPPORTED_ONLY` “carrier could not be identified” `BOTH`
29. S5 `SUPPORTED_ONLY` R3 1 - `ABSTAIN` `R3.G.SIGNIFICANCE` IRM magnetite 1
30. S6

1

S1-S6 R3/R5/R7 S4

2

3

“ ”

4

`scope_id` /

5

`SUPPORT` `CONFLICT` `UNKNOWN` `ACTIVE/DEFEATED/UNDECIDED`

6 R3/R5/R7

R3 R5 R7 A95

7

1 0 `ABSTAIN` `BOTH` 0

8

31. S1-S6
32.
33.
34.
35. `scope_id`
36.
37. `ACTIVE/DEFEATED/UNDECIDED`
38. 1/0/`ABSTAIN`
39. R3 -

40. R5
41. R7 A95 R
42.
43.
44. S1-S6
45.

S1 `criterion_spec.py` `expert_conflict_engine.py` S2 MinerU S3 `paper_text_retrieval.py` `r7_original_text_extractor.py` S4 `expert_conflict_engine.py` `r3_rockmag_distilled_reasoner.py` `r5_r7_expert_rules.py` `r7_pole_overlap_reasoner.py` S5 R3 R5 R7 S6 worker

R3 V3 `REFNO` OOF 785/872 90.02% 100% 2316/2751 84.19% GREP 2325/2751 84.51% `REFNO` bootstrap 0

46. R3 R5 R7 90%
47. strict workflow
48. R7
49. OOF
50. SHA-256
51. Python MinerU JSONL XLSX
52. `ABSTAIN`
53.
54. PDF
55. `scope_id` “ ” “ ”
56.
57. `ABSTAIN` 0 1
58.

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